LA SPRING 2026

MID A

GRADING DECISIONS

Q 1. (a) The given equation is NOT true. This can be shown with **any** correct counterexample. The simplest one would be to use two 2 x 2 matrices, such as the case below to disprove the result:

EXAMPLE: FOR $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, $A+B = \begin{bmatrix} 4 & 3 \\ 3 & 8 \end{bmatrix}$

VERIFY (P.93 7th ED) $\boxed{\det(A) + \det(B) \neq \det(A+B)}$ P.9 8th

HINT: $\det(A) = 1$, $\det(B) = 8$, $\det(A+B) = 23$, $9 \neq 23$

(b)

Solution: This follows from Theorem 2.1.2 and the fact that the cofactors of A are integers if A has only integer entries, since integers are closed under multiplication, addition and subtraction. $A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) = \operatorname{adj}(A)$

MARKING SCHEME

Q. 2

Proof Let R be the reduced row-echelon form of A. As a preliminary step, we will show that $\det(A)$ and $\det(R)$ are both zero or both nonzero: Let $E_1, E_2, \cdots, E_r$ be the elementary matrices that correspond to the elementary row operations that produce R from A. Thus

$$R = E_r \cdots E_2 E_1 A$$

and from 3,

$$\det(R) = \det(E_r) \cdots \det(E_2) \det(E_1) \det(A) \tag{4}$$

But from Theorem 2.2.4 the determinants of the elementary matrices are all nonzero. (Keep in mind that multiplying a row by zero is *not* an allowable elementary row operation, so $k \neq 0$ in this application of Theorem 2.2.4.) Thus, it follows from 4 that $\det(A)$ and $\det(R)$ are both zero or both nonzero. Now to the main body of the proof.

If A is invertible, then by Theorem 1.6.4 we have $R = I$, so $\det(R) = 1 \neq 0$ and consequently $\det(A) \neq 0$. Conversely, if $\det(A) \neq 0$, then $\det(R) \neq 0$, so R cannot have a row of zeros. It follows from Theorem 1.4.3 that $R = I$, so A is invertible by Theorem 1.6.4.

THEOREM 1.4.3

If R is the reduced row-echelon form of an $n \times n$ matrix A, then either R has a row of zeros or R is the identity matrix I_n.

MARKING SCHEME

Q. 3

$$\begin{aligned} x + 2y &= 4 \\ 3x + ay &= 6 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & | & 4 \\ 3 & a & | & 6 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 3R_1$$
$$\begin{bmatrix} 1 & 2 & | & 4 \\ 0 & a-6 & | & -6 \end{bmatrix}$$

Case 1: Exactly one solution

Occurs when the pivot is nonzero:

$$\mathrm{a} - 6 \neq 0$$
$$\boxed{\mathrm{a} \neq 6}$$

Because both variables have pivots → unique solution.

Case 2: No solution

Occurs when:

$$\mathrm{a} - 6 = 0$$

Substitute $\mathrm{a} = 6$:

Second row becomes:

$$0\mathrm{y} = -6$$

This is impossible → inconsistent.

$$\boxed{\mathrm{a} = 6 \Rightarrow \text{no solution}}$$

Case 3: Infinitely many solutions

This requires:

$$0\mathrm{y} = 0$$

But the right side is always -6, never 0.

So this never happens.

$$\boxed{\text{No value of a}}$$

MARKING SCHEME

Q. 4 Since A is invertible, there exists a matrix A^{-1} such that

$$A A^{-1} = A^{-1} A = I.$$

Take the transpose of both sides of the equation $A A^{-1} = I$:

$$\left(A A^{-1}\right)^T = I^T.$$

Using the properties of transpose,

$$\left(A A^{-1}\right)^T = \left(A^{-1}\right)^T A^T \text{ and } I^T = I.$$

Thus,

$$\left(A^{-1}\right)^T A^T = I.$$

Similarly, taking the transpose of $A^{-1} A = I$ gives

$$A^T \left(A^{-1}\right)^T = I.$$

Therefore, $\left(A^{-1}\right)^T$ is both a left and right inverse of A^T.
By uniqueness of the inverse, we conclude that

$$\left(A^T\right)^{-1} = \left(A^{-1}\right)^T.$$

MARKING SCHEME